Discover the Many Benefits of the Redis Database

Today, apart from relational tables, data is also available in other forms such as images, texts, blogs, lists, etc. Redis can be used as a database, cache and message broker. It is exceptionally fast, and can support five data types and two special types of data.

Redis is an open source, in-memory and key/value NoSQL database. Known to be an exceptionally fast in-memory database, Redis is used as a database, cache and message broker, and is written in C. It supports five data types— strings, hashes, lists, sets, sorted sets and two special types of data—Bitmap and HyperLogLog.

Since Redis runs in memory, it is very fast but is disk persistent. So in case a crash happens, data is not lost. Redis can perform about 110,000 SETs and about 81,000 GETs per second. This popular database is used by many companies such as Pinterest, Instagram, StackOverflow, Docker, etc.

There are several problems that Redis addresses, which are listed below.

1. *Session cache:* Redis can be used to cache user sessions. Due to its in-memory data structure as well as data persistent nature, Redis is a prefect choice for this use case.
2. *Middle layer:* If your program delivers data at a very fast rate but the process of storing data in a database such as MySQL is very slow, then you can use Redis as the middle layer. Your program will just put the data into Redis, and another program will pull the data and store it in the database.
3. *Shows the latest items listed in your home page:* Due to fast retrieval, you can show the important data to users from Redis. The rest of the data can be fetched from the database later.
4. It acts as a broker for Celery processes.

Installation of Redis

You can download a stable version of Redis from *https://redis.io/download*. We are going to install Redis on Centos 6.7 —you can use Ubuntu or Debian. Let us extract the tar package of Redis, for which you can use the following command:

```
$ tar -xvzf redis-3.2.8.tar.gz
```

After extraction, a directory *redis-3.2.8* will be formed. Just use one command, *make*, to install the Redis. Sometimes, an error occurs while installing Redis:

```
In file included from adlist.c:34:
zmalloc.h:50:31: error: jemalloc/jemalloc.h: No such file or
directory
zmalloc.h:55:2: error: #error "Newer version of jemalloc
required"
make[1]: *** [adlist.o] Error 1
make[1]: Leaving directory `/root/redis/redis-3.2.8/src'
```